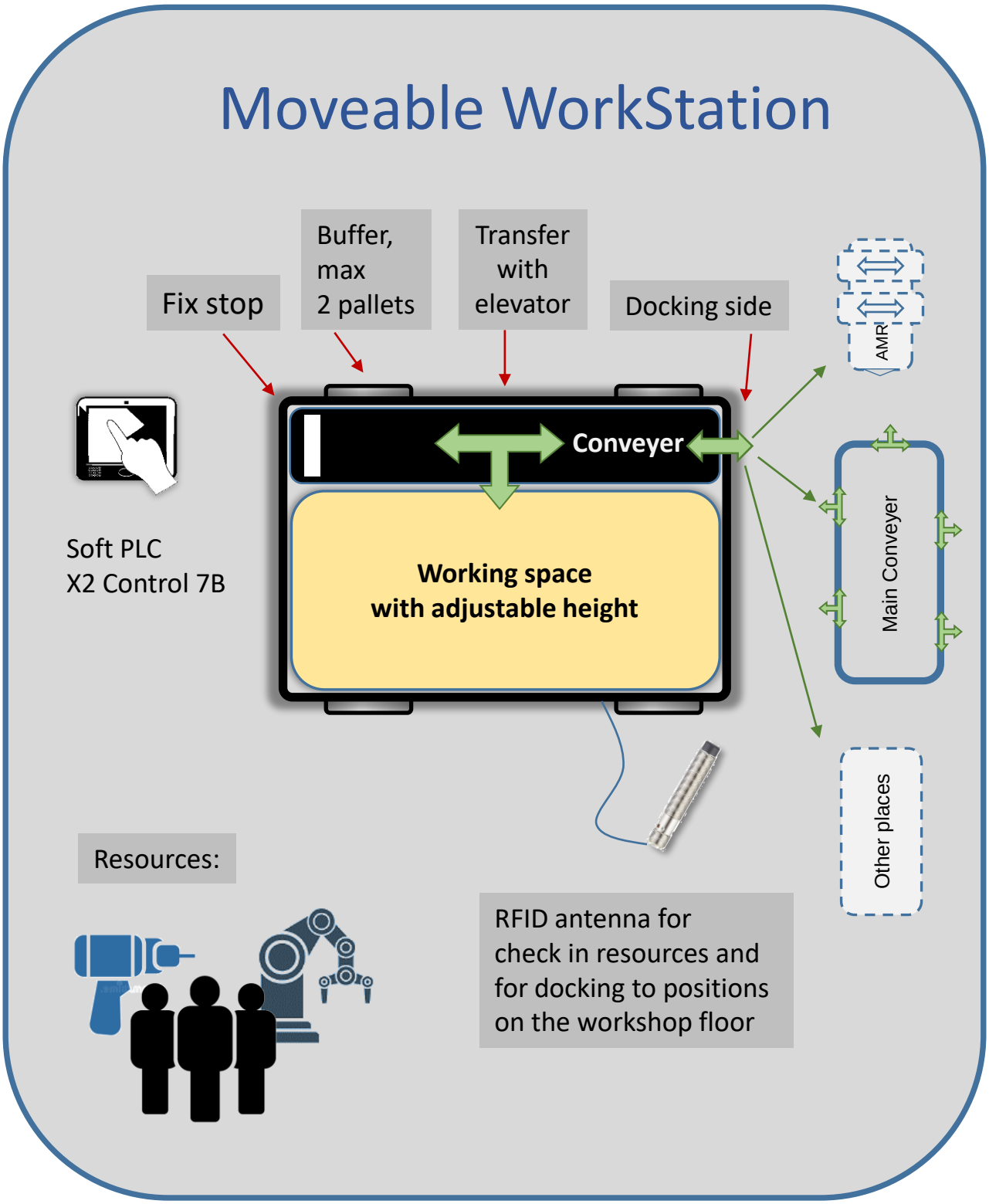
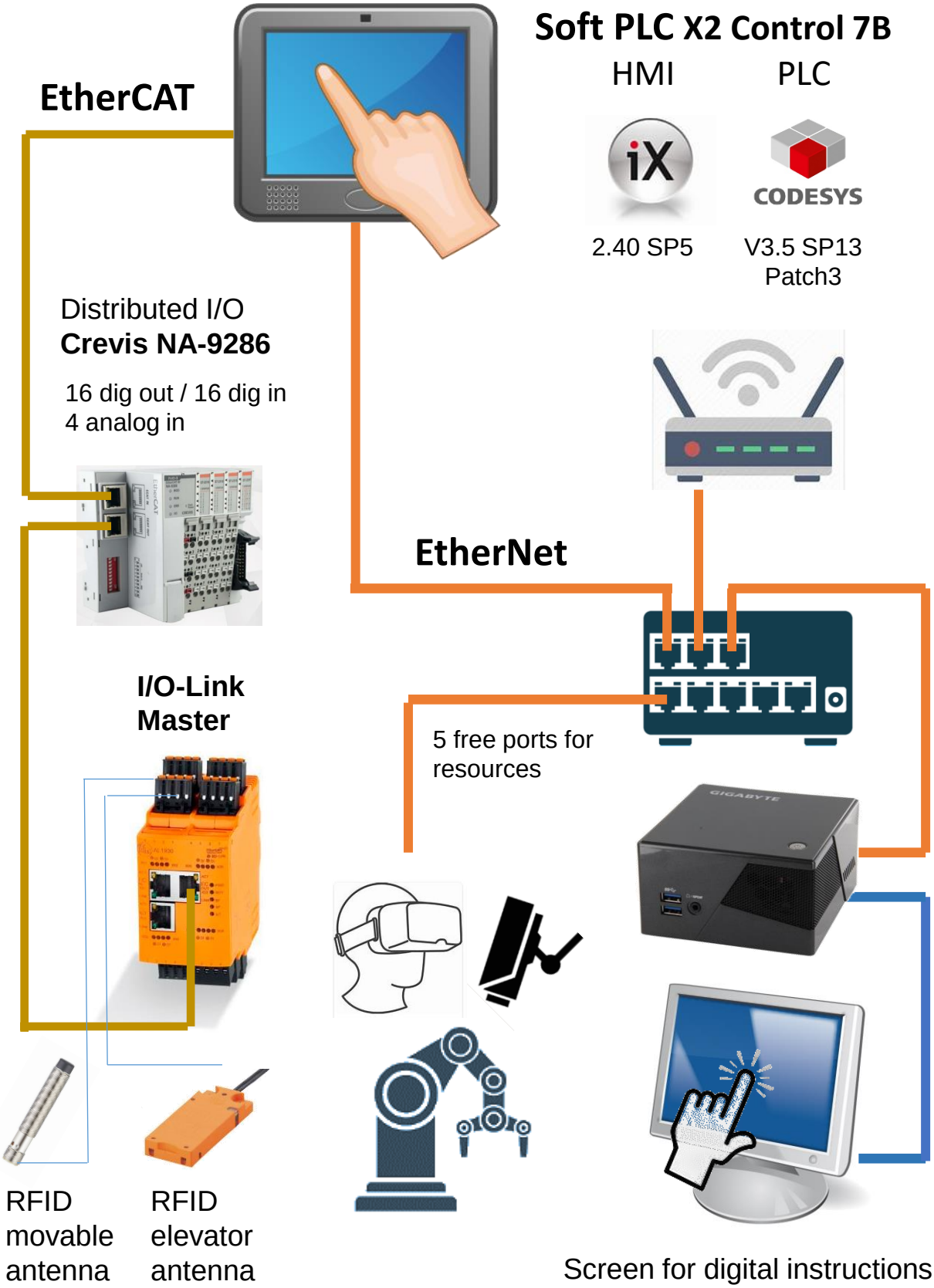
Program code

WS Conveyor SII-Lab



Connected item at WS



WS1-WS3 I/O functions

pallet_sensor1

pallet_sensor2

buffer_queue_
stop_sensor

buffer_queue_
stop_down

4

14

5 1 3

12

elevator_sensor_up

elevator_sensor_down

table_height_sensor

M

transfer_
dir_from_WSpace

transfer_on

M

coveyor_
dir_to_ws

conveyor_
on

To table height remote control

4 2

14

5 1 3

12

elevator_down

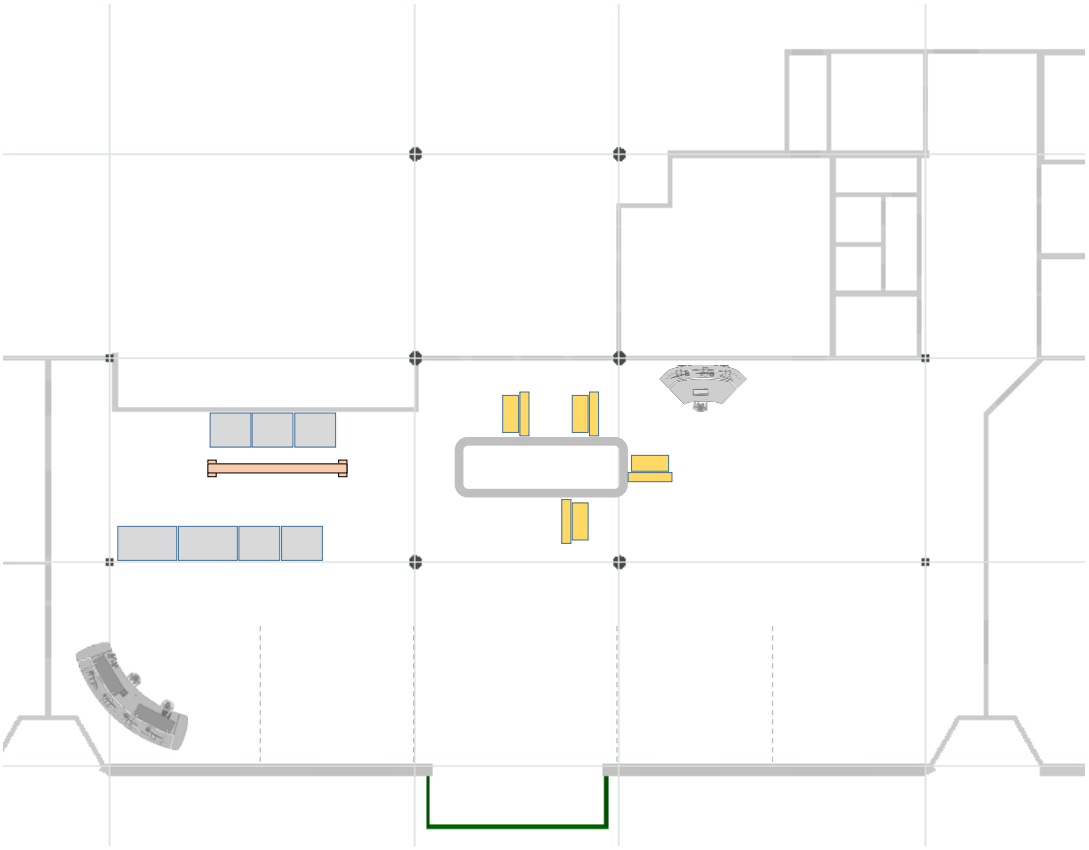
elevator_up

table_up

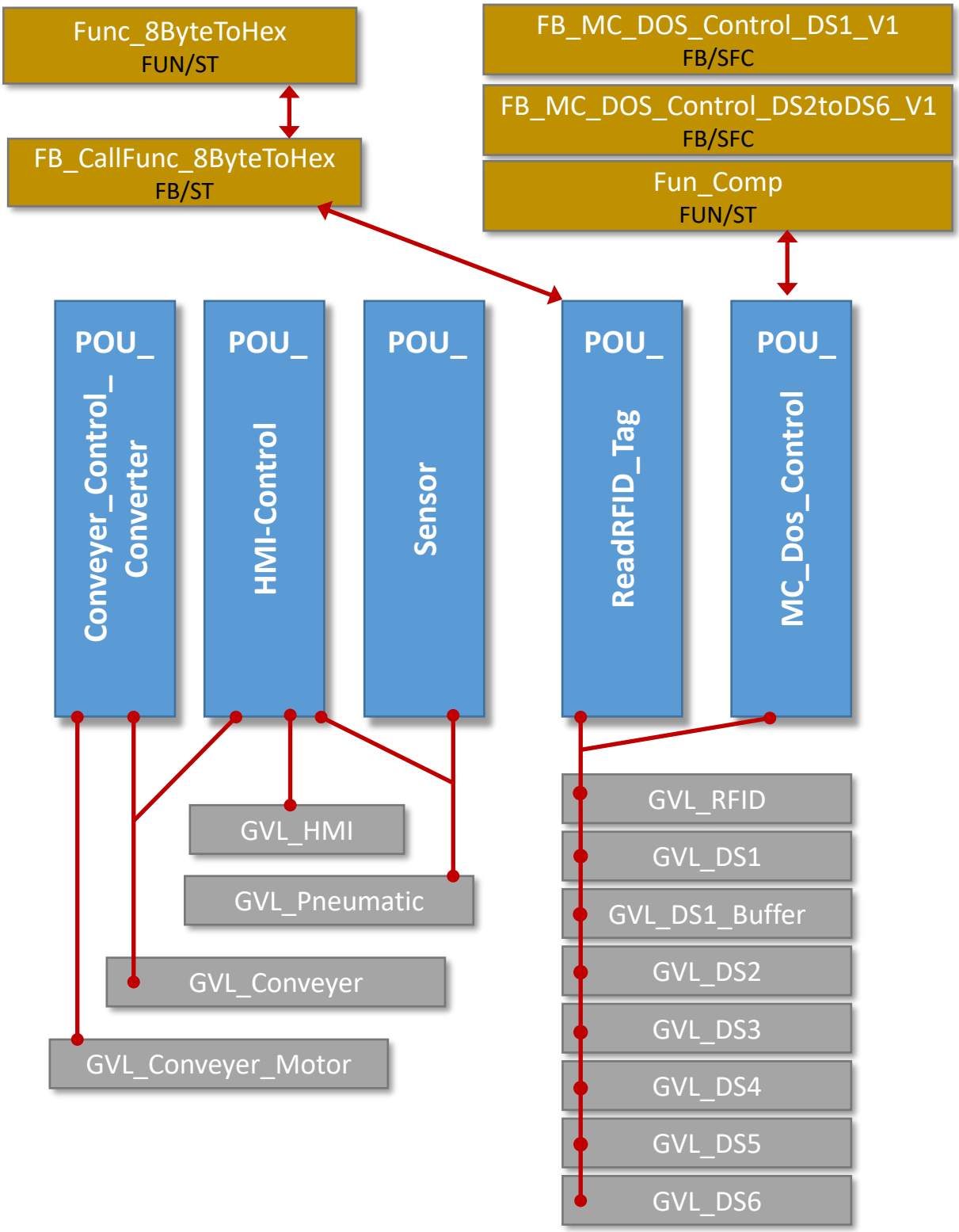
table_down

Crevis NA-9286

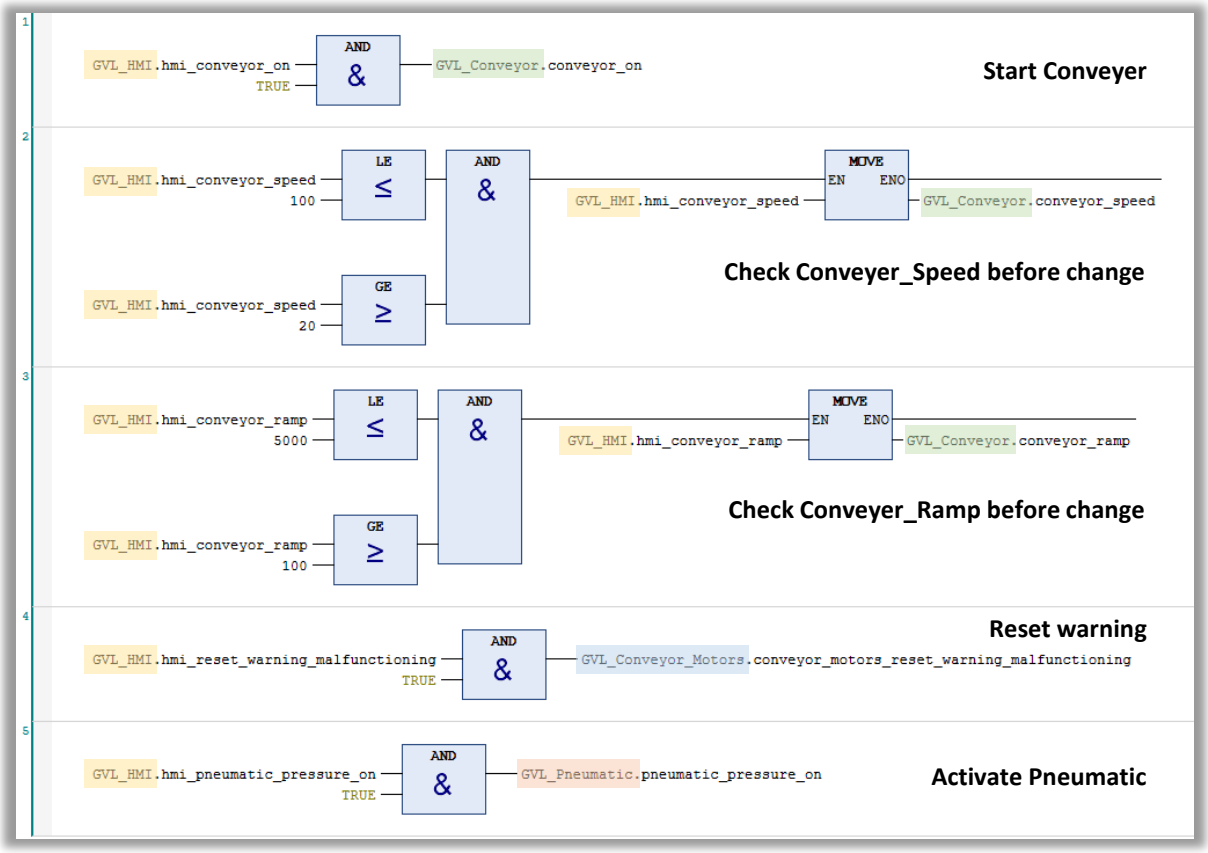
Variable	Channel	Address	Type
NA_9286_WS1			
ST_1218_1			
	Byte 0	%IB0	USINT
Application.GVL_WS1_IO.buffer_queue_stop_sensor	Bit0	~~%IX0.0~~	BOOL
Application.GVL_WS1_IO.pallet_sensor_2	Bit1	~~%IX0.1~~	BOOL
Application.GVL_WS1_IO.pallet_sensor_1	Bit2	~~%IX0.2~~	BOOL
Application.GVL_WS1_IO.elevator_sensor_up	Bit3	~~%IX0.3~~	BOOL
Application.GVL_WS1_IO.elevator_sensor_down	Bit4	~~%IX0.4~~	BOOL
	Bit5	%IX0.5	BOOL
ST_1218			
	Byte 0	%IB1	USINT
ST_2328_1			
	Byte 0	%QB0	USINT
Application.GVL_WS1_IO.buffer_queue_stop_down	Bit0	~~%QX0.0~~	BOOL
Application.GVL_WS1_IO.elevator_down	Bit1	~~%QX0.1~~	BOOL
Application.GVL_WS1_IO.elevator_up	Bit2	~~%QX0.2~~	BOOL
Application.GVL_WS1_IO.coveyor_on	Bit3	~~%QX0.3~~	BOOL
Application.GVL_WS1_IO.conveyor_dir_to_WStation	Bit4	~~%QX0.4~~	BOOL
Application.GVL_WS1_IO.transfer_on	Bit5	~~%QX0.5~~	BOOL
Application.GVL_WS1_IO.transfer_dir_from_WSpace	Bit6	~~%QX0.6~~	BOOL
	Bit7	%QX0.7	BOOL
ST_2328			
	Byte 0	%QB1	USINT
Application.GVL_WS1_IO.table_down	Bit0	~~%QX1.0~~	BOOL
Application.GVL_WS1_IO.table_up	Bit1	~~%QX1.1~~	BOOL
	Bit2	%QX1.2	BOOL
ST_3424			
Application.GVL_WS1_IO.table_height_sensor	Analog Input Ch#0 Byte	~~%IW1~~	UINT
	Analog Input Ch#1 Byte	%IW2	UINT



Project Structue
CoDeSys V3.5 SP13 Patch3

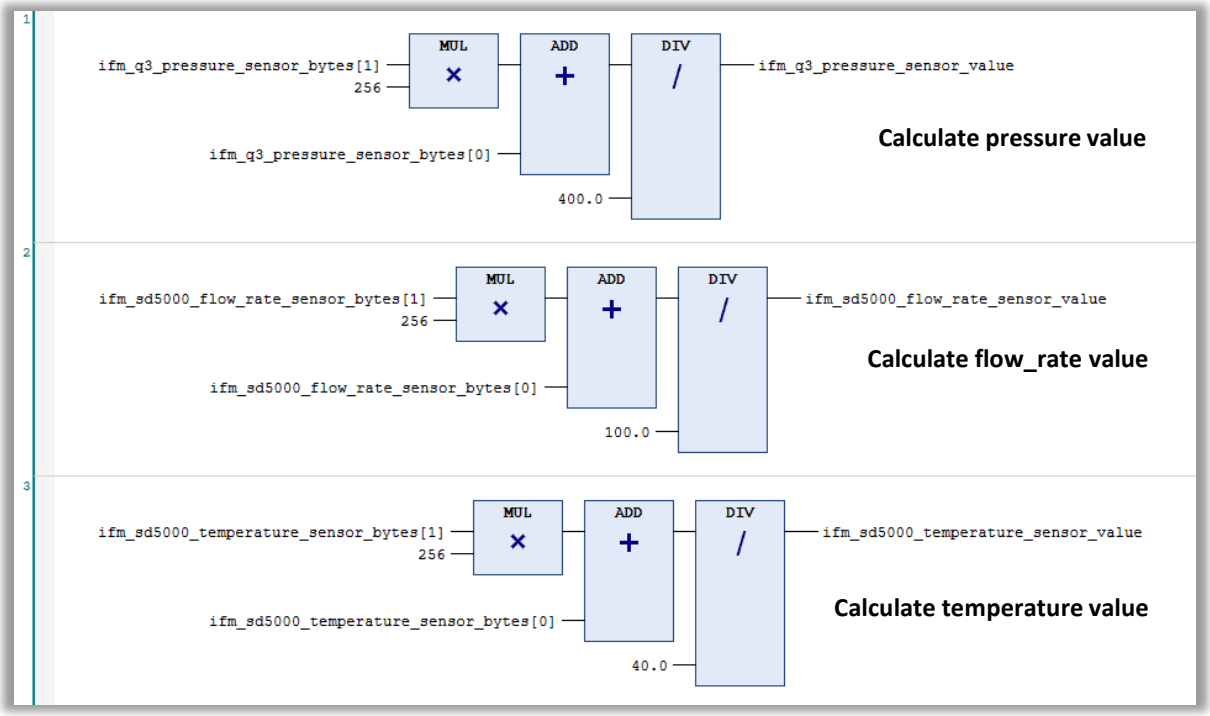


POU_HMI_Control



POU_Sensors

All variables are in GVL_Pneumatic !!



POU_ReadRFID_Tag

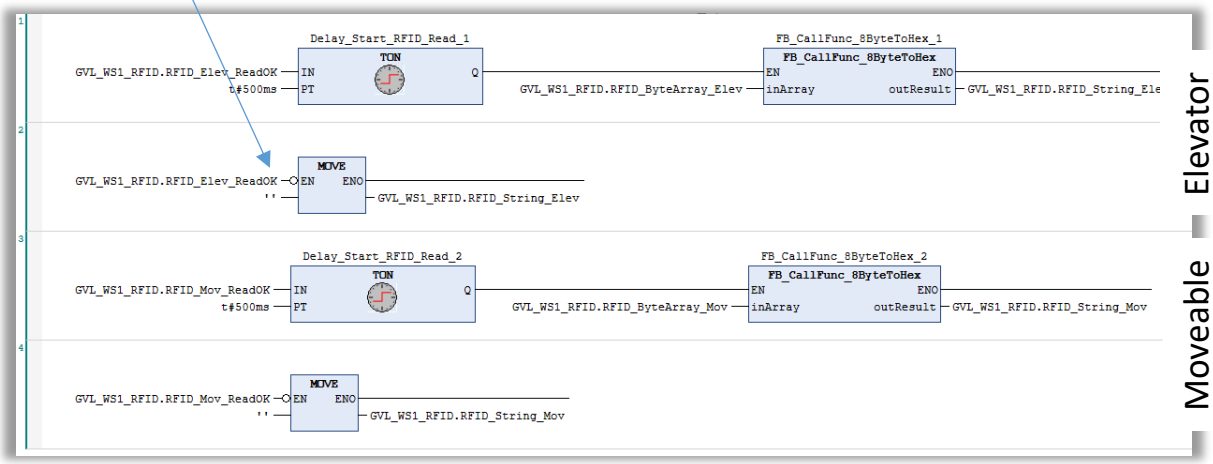
Scope	Name	Address	Data type
VAR	**FB_CallFunc_8ByteToHex_1**		FB_CallFunc_8ByteToHex
VAR	**Delay_Start_RFID_Read_1**		TON
VAR	**FB_CallFunc_8ByteToHex_2**		FB_CallFunc_8ByteToHex
VAR	**Delay_Start_RFID_Read_2**		TON

Local variable list

When a pallet reached queue_stop wait 0.5 s before start converting result from RFID antenna

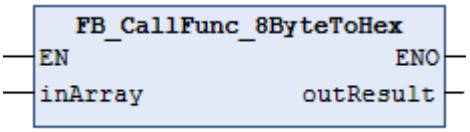
Reset RFID_string when pallet leave queue_stop

Call function for converting byte_arry to String



FB_CallFunc_8ByteToHex

Function Block



Local variable list

Scope	Name	Address	Data type	Initialization
VAR_INPUT	**inArray**		ARRAY[0..7] OF BYTE	
VAR_OUTPUT	**outResult**		STRING	''

Program Code (ST)

```
outResult := '';
outResult := Func_8ByteToHex(inArray) ;
```

Call function for converting byte_arry to String

Func_8ByteToHex

Local variable list

	Scope	Name	Address	Data type	Initialization
1	VAR_INPUT	**InArray**		ARRAY[0..7] OF BYTE	
2	VAR	**myHexLista**		STRING	'0123456789ABCDEF'
3	VAR	**myString**		STRING	
4	VAR	**myLoopVar**		USINT	
5	VAR	**myNib1**		USINT	
6	VAR	**myNib2**		USINT	
7	VAR	**myChar1**		STRING	
8	VAR	**myChar2**		STRING	

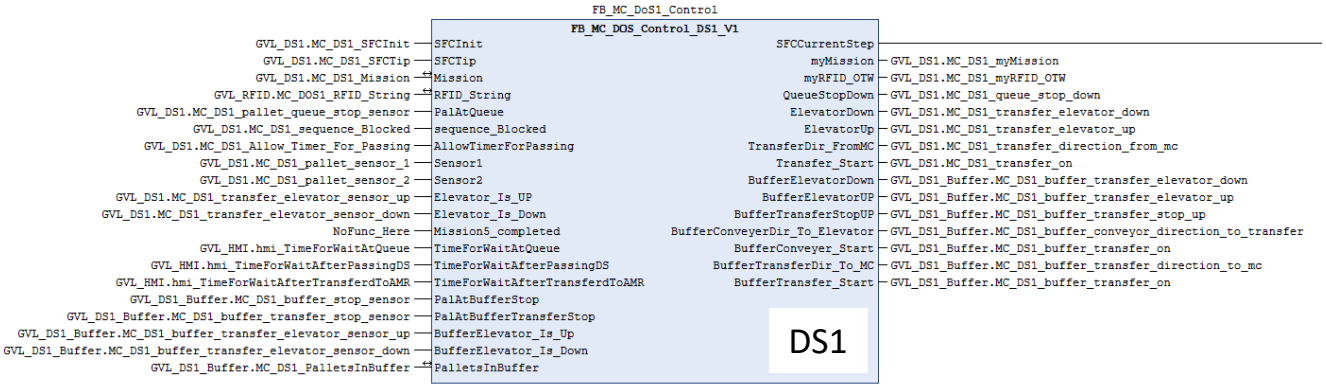
Program Code (ST)

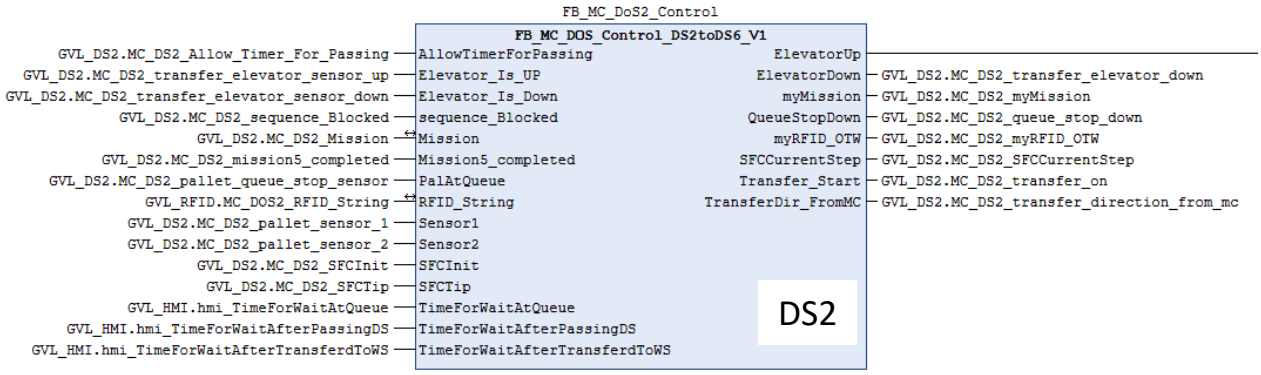
```
1   myString:= '';
2
3   FOR myLoopVar:=0 TO 7 DO
4       myNib2 := InArray[myLoopVar] AND 15;
5       myNib1 := SHR (InArray[myLoopVar],4);
6       myChar1 := MID (myHexLista,1,myNib1+1);
7       myChar2 := MID (myHexLista,1,myNib2+1);
8       myString := CONCAT ( myString,myChar1);
9       myString := CONCAT ( myString,myChar2);
10  END_FOR
11
12  Func_8ByteToHex := myString;
```

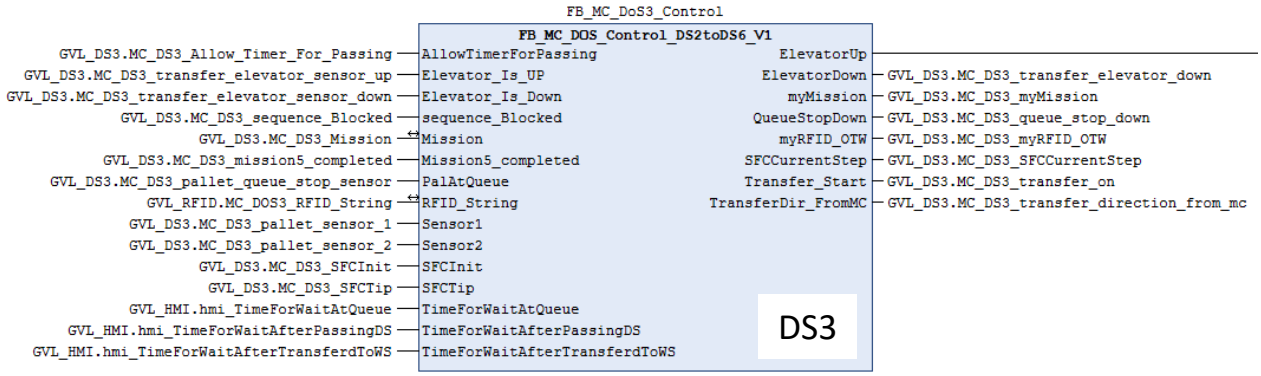
POU_MC_Dos_Control 1(2)

Local variable list

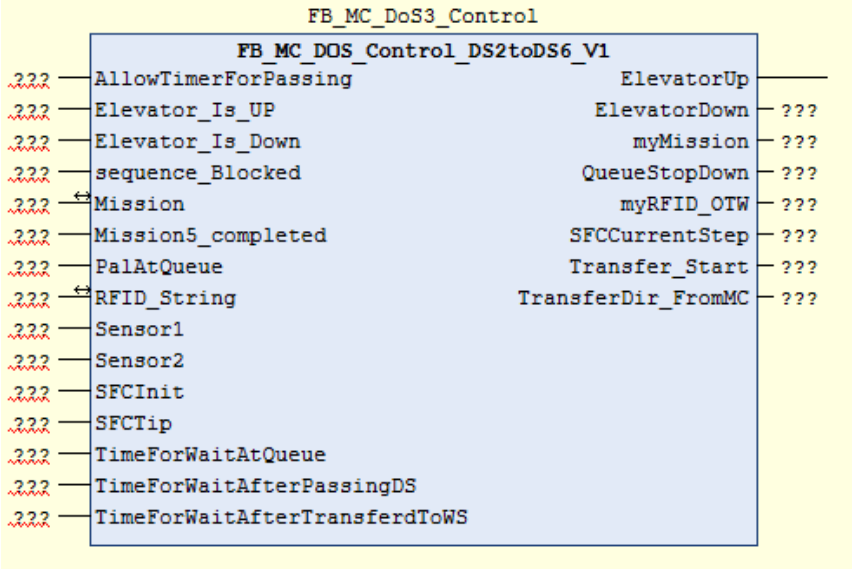
	Scope	Name	Address	Data type
1	VAR	**FB_MC_DoS1_Control**		FB_MC_DOS_Control_DS1_V1
2	VAR	**FB_MC_DoS2_Control**		FB_MC_DOS_Control_DS2toDS6_V1
3	VAR	**FB_MC_DoS3_Control**		FB_MC_DOS_Control_DS2toDS6_V1
4	VAR	**FB_MC_DoS4_Control**		FB_MC_DOS_Control_DS2toDS6_V1
5	VAR	**FB_MC_DoS5_Control**		FB_MC_DOS_Control_DS2toDS6_V1
6	VAR	**FB_MC_DoS6_Control**		FB_MC_DOS_Control_DS2toDS6_V1
7	VAR	**NoFunc_Here**		BOOL





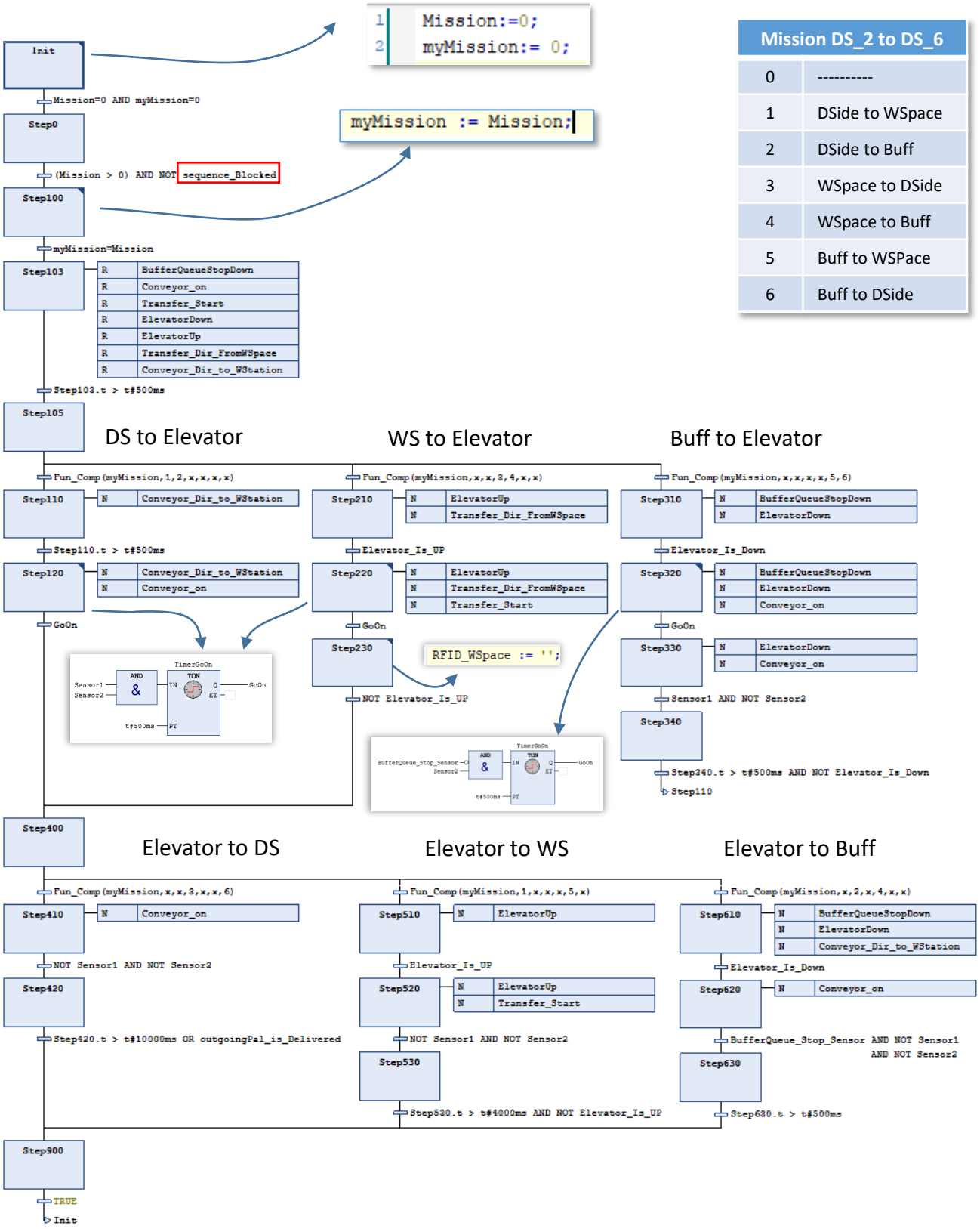


DS2 – DS6
Functionsblock



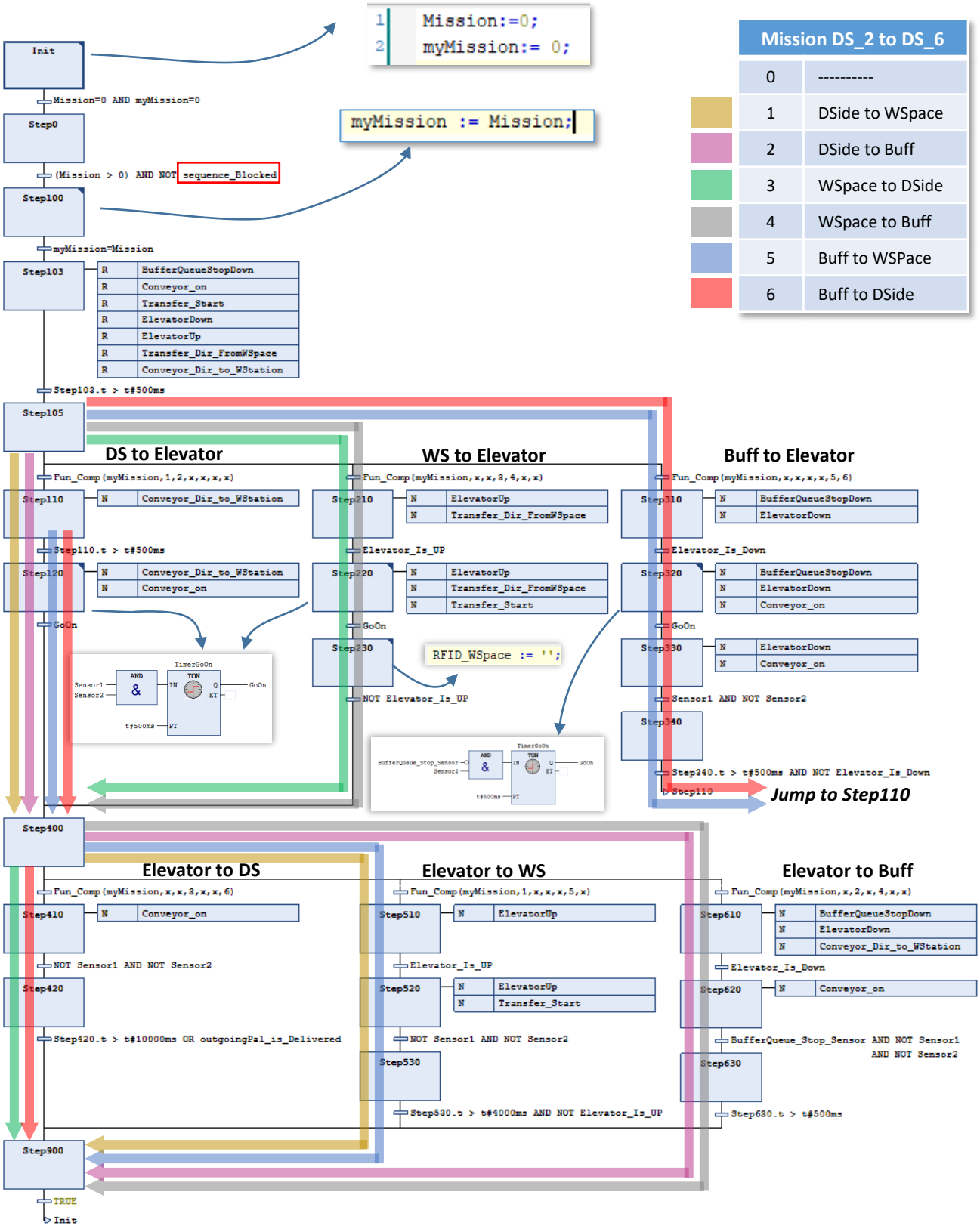
	Scope	Name	Address	Data type	Initialization	Comment
1	VAR_INPUT	**AllowTimerForPassing**		BOOL		Tag for activate timer
2	VAR_OUTPUT	**ElevatorUp**		BOOL		
3	VAR_OUTPUT	**ElevatorDown**		BOOL		
4	VAR_INPUT	**Elevator_Is_UP**		BOOL		
5	VAR_INPUT	**Elevator_Is_Down**		BOOL		
6	VAR_INPUT	**sequence_Blocked**		BOOL		For stoping Mission to start a sequece
7	VAR_IN_OUT	**Mission**		USINT		Mission for DS
8	VAR_OUTPUT	**myMission**		USINT	0	Stored Mission during Sequence
9	VAR_INPUT	**Mission5_completed**		BOOL		PushButton_activated
10	VAR_INPUT	**PalAtQueue**		BOOL		Pallet att queue
11	VAR_OUTPUT	**QueueStopDown**		BOOL		
12	VAR_IN_OUT	**RFID_String**		STRING		Result from RFID reader
13	VAR_OUTPUT	**myRFID_OTW**		STRING	"	Hold result from RFID reader when leaving queue
14	VAR_INPUT	**Sensor1**		BOOL		
15	VAR_INPUT	**Sensor2**		BOOL		
16	VAR_OUTPUT	**SFCCurrentStep**		STRING		Shows current step name
17	VAR_INPUT	**SFCInit**		BOOL		Reset sequence to the initialk step
18	VAR_INPUT	**SFCTip**		BOOL		Jogging sequence to next step
19	VAR	**TimerForPassing**		TON		Timer for passing
20	VAR_INPUT	**TimeForWaitAtQueue**		TIME		Time for waiting, at queue stop, before mission 1 is activated
21	VAR_INPUT	**TimeForWaitAfterPassingDS**		TIME		Time for waiting after passing Elevator to MC
22	VAR_INPUT	**TimeForWaitAfterTransferdToWS**		TIME		Time for wating after transferig to WS
23	VAR_OUTPUT	**Transfer_Start**		BOOL		
24	VAR_OUTPUT	**TransferDir_FromMC**		BOOL		Tag for aktivate Dir_FromMC
25	VAR CONSTANT	**x**		INT	-1	

WS Program sequence inside funktionsblock

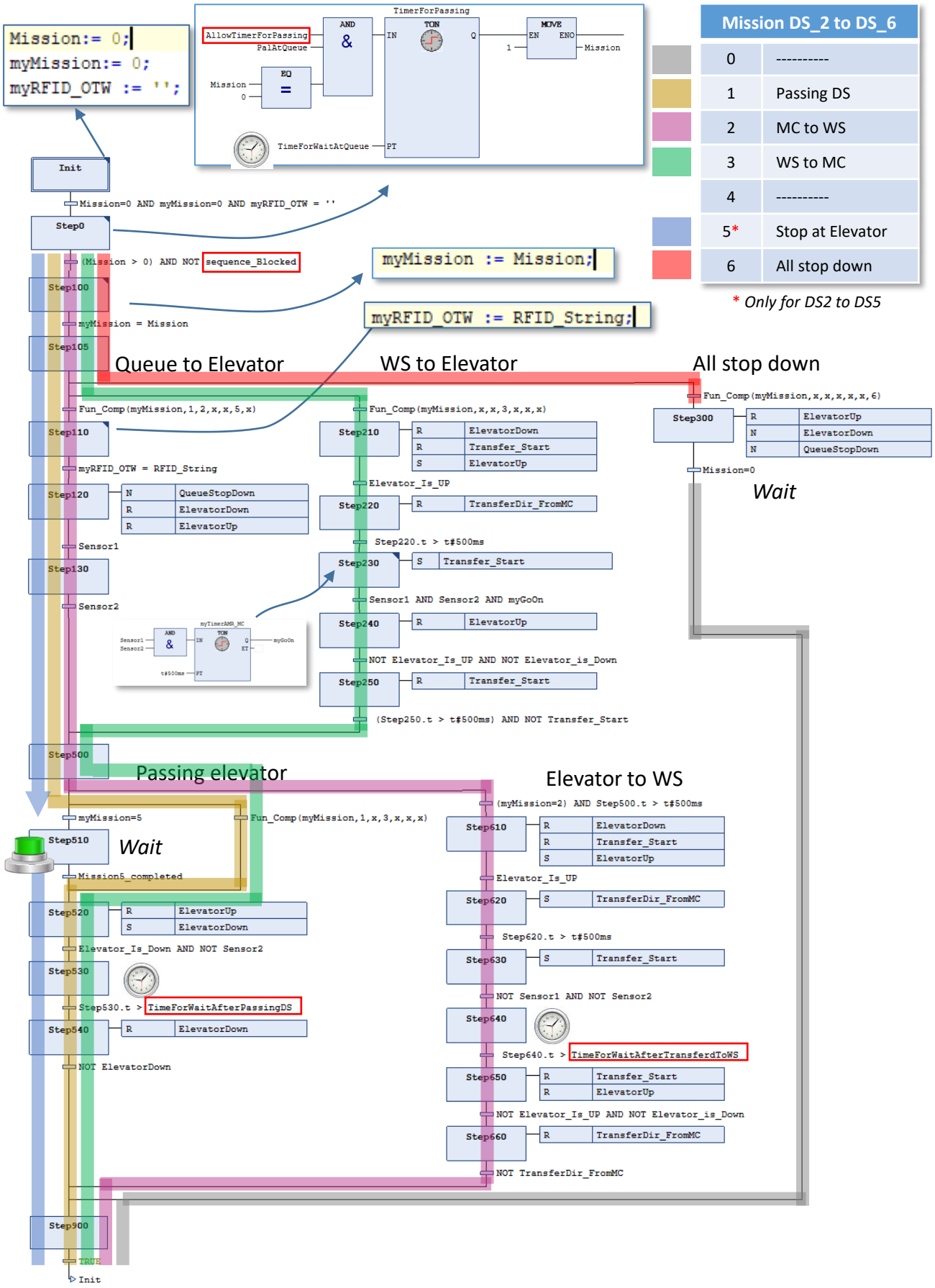


Mission DS_2 to DS_6	
0	----------
1	DSide to WSpace
2	DSide to Buff
3	WSpace to DSide
4	WSpace to Buff
5	Buff to WSPace
6	Buff to DSide

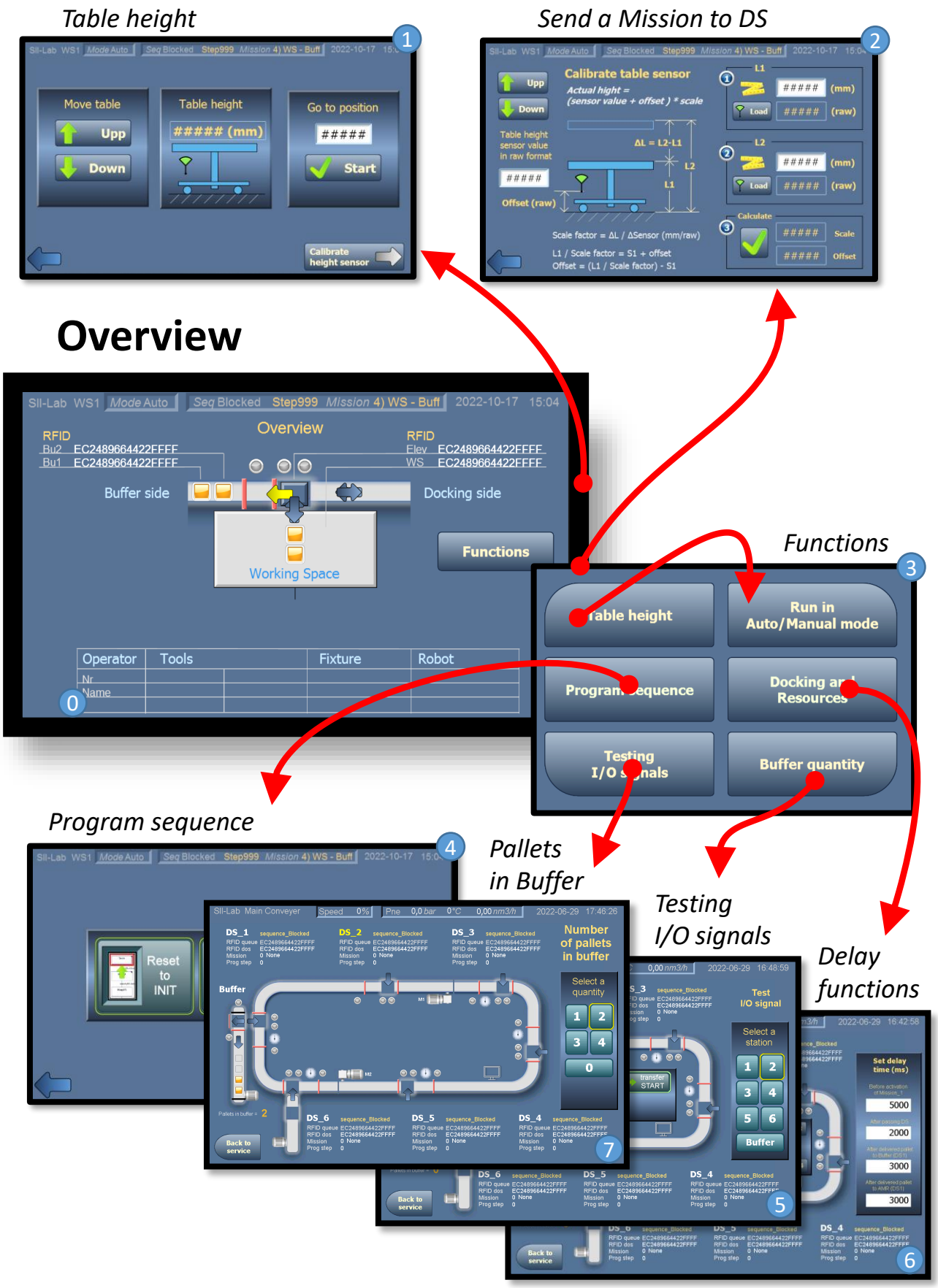
WS Program sequence inside funktionsblock



DS2 – DS6 Mission sequence inside funktionsblock

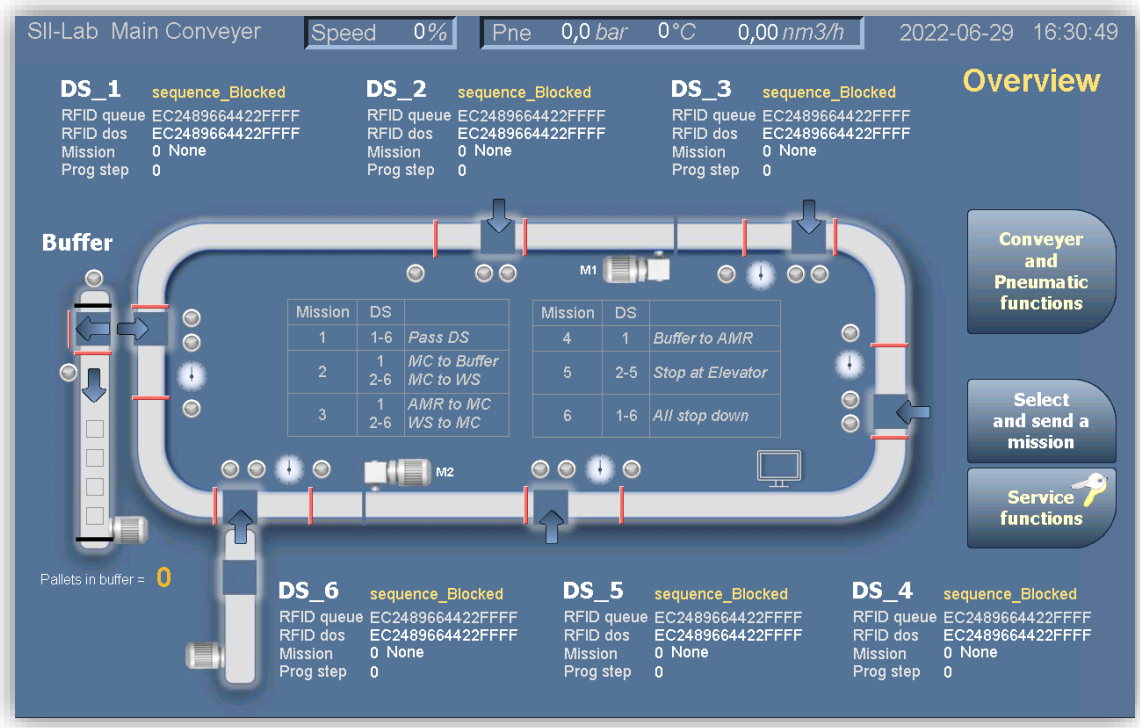


HMI screens WS

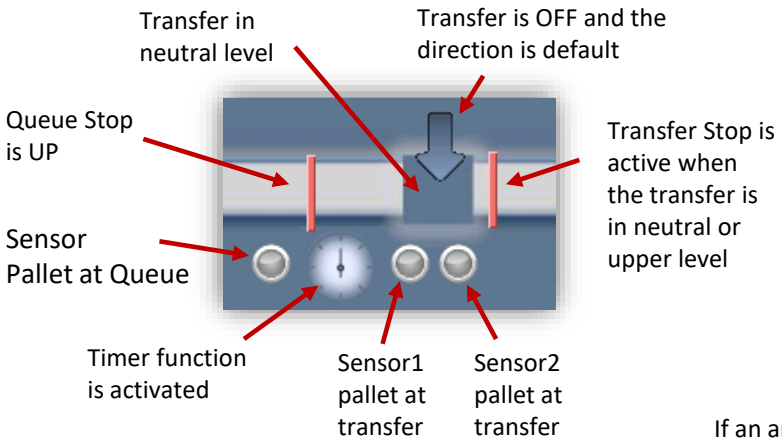


HMI screen

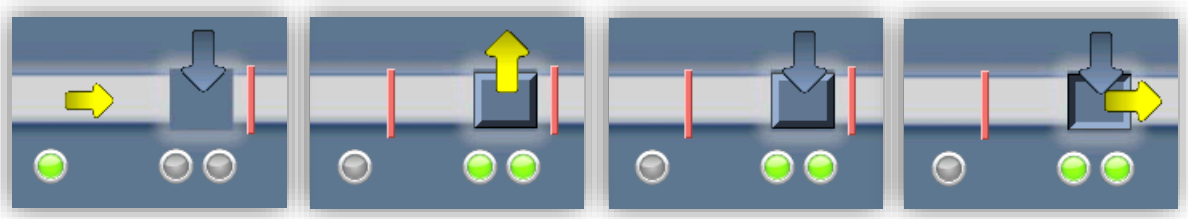
Overview 1(3)



Explanation of Conveyer Icon's



If an arrow is yellow the conveyor belt is running



Queue Stop is down

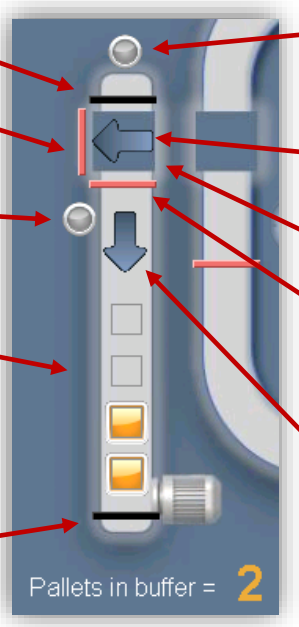
Transfer is in upper level. It is ON, and the direction is ToWS

Transfer is in upper level. It is OFF, and the direction is default

Transfer is in lower level. It is OFF, and the direction is default

HMI screen
Overview 2(3)

Explanation of Buffer Icon's



Fix STOP

Queue Stop is UP

Sensor Pallet at Queue

Number of pallets in buffer

Fix STOP

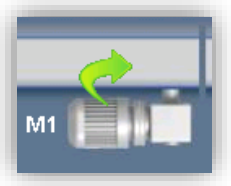
Sensor Pallet on Buffer Transfer

Transfer is OFF and the direction is default

Transfer elevator is in neutral level

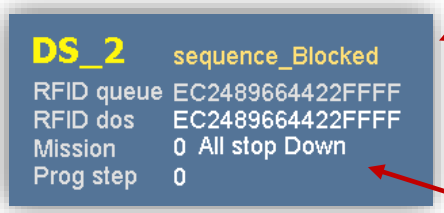
Transfer Stop is active when the transfer is in neutral or upper level

Buffer conveyer is OFF and the direction is default



The green arrow indicate that the main conveyor is running

Explanation of DS information text



DS group number. The color indicate if the station is selected or not

If a pallet stopped at queue the tag information shows here

When a pallet got a mission and leave the queue position the tag information from queue stores here. This information is available until the mission ends.

Shows active STEP in the SFC sequence

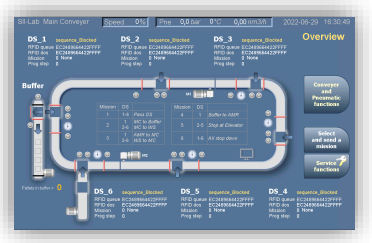
This text is visible if you blocked the program sequence (SFC).

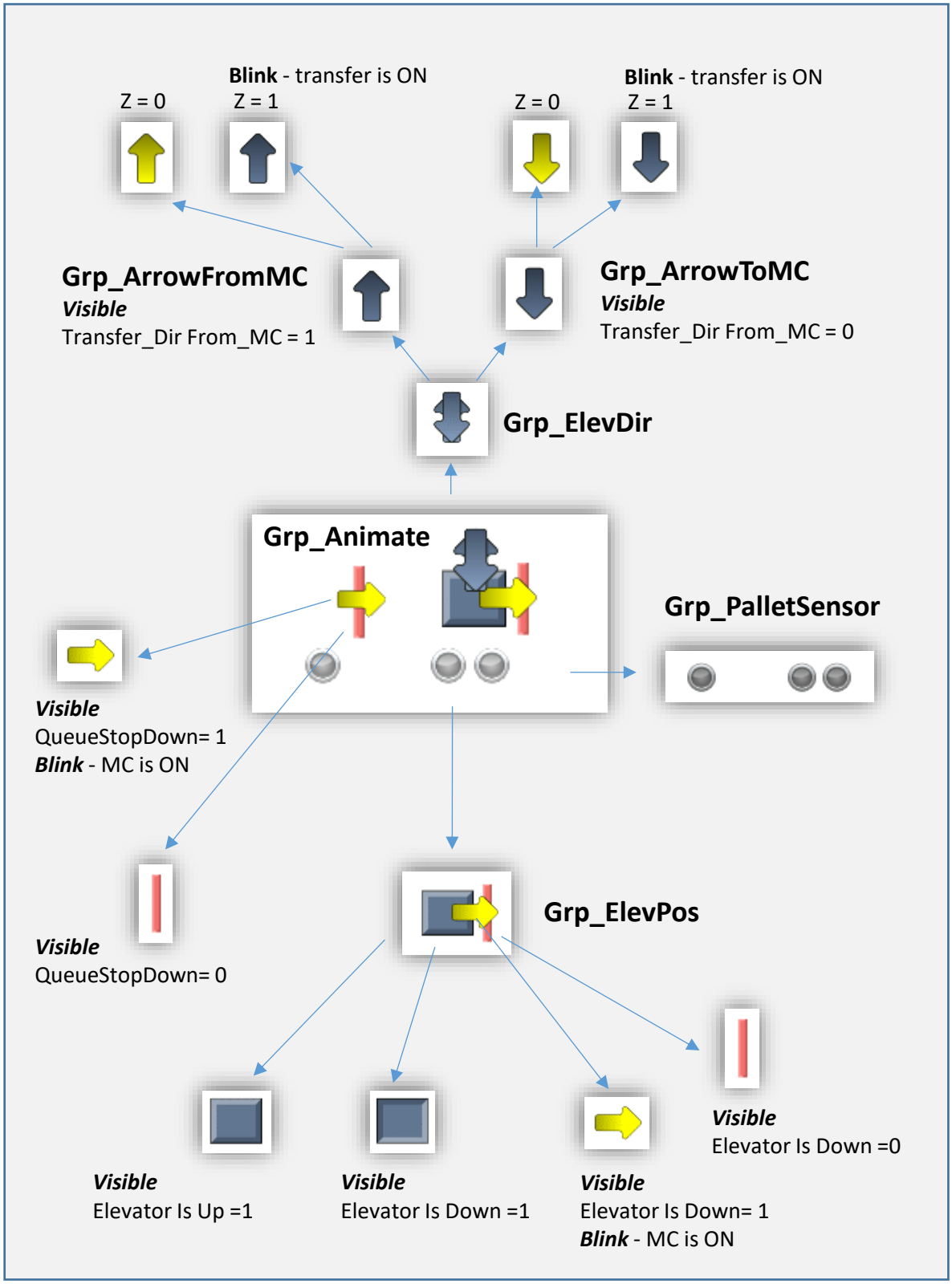
Information about active mission in the SFC sequence

HMI screen

Overview 3(3)

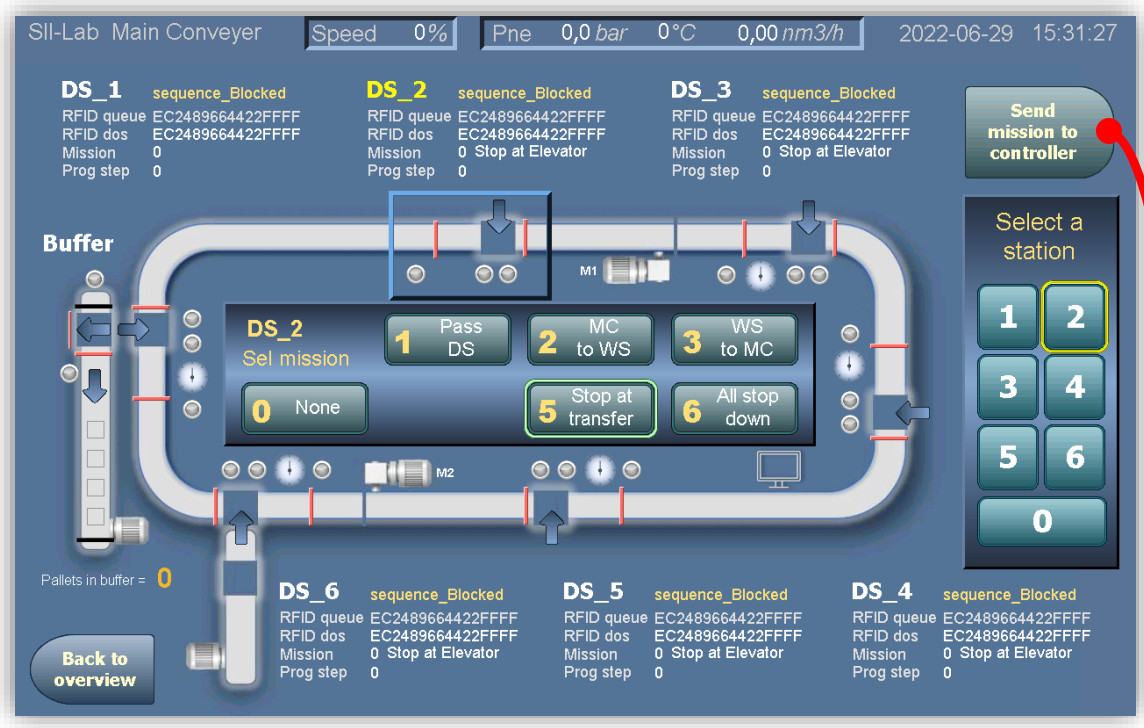
Explanation of Icon's code





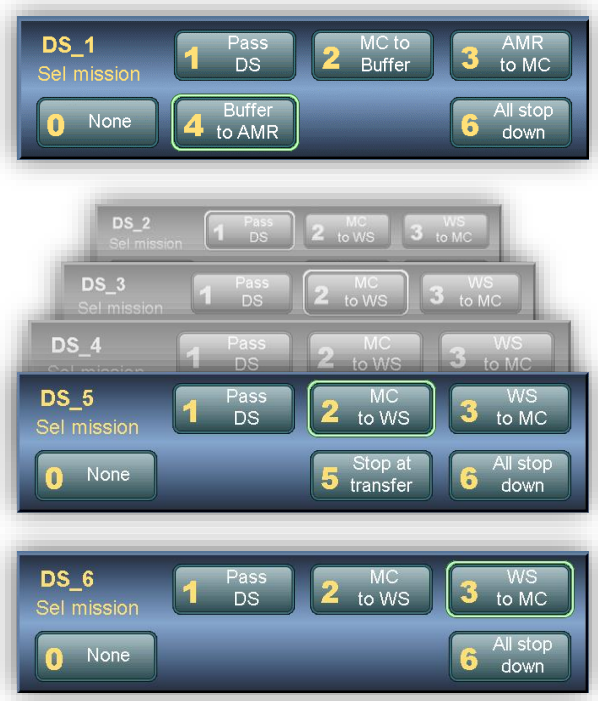
HMI screen

Send a Mission



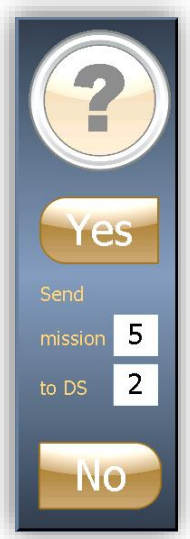
Available Missions fore DS1 – DS6

A green frame around the button indicates the Mission you selected



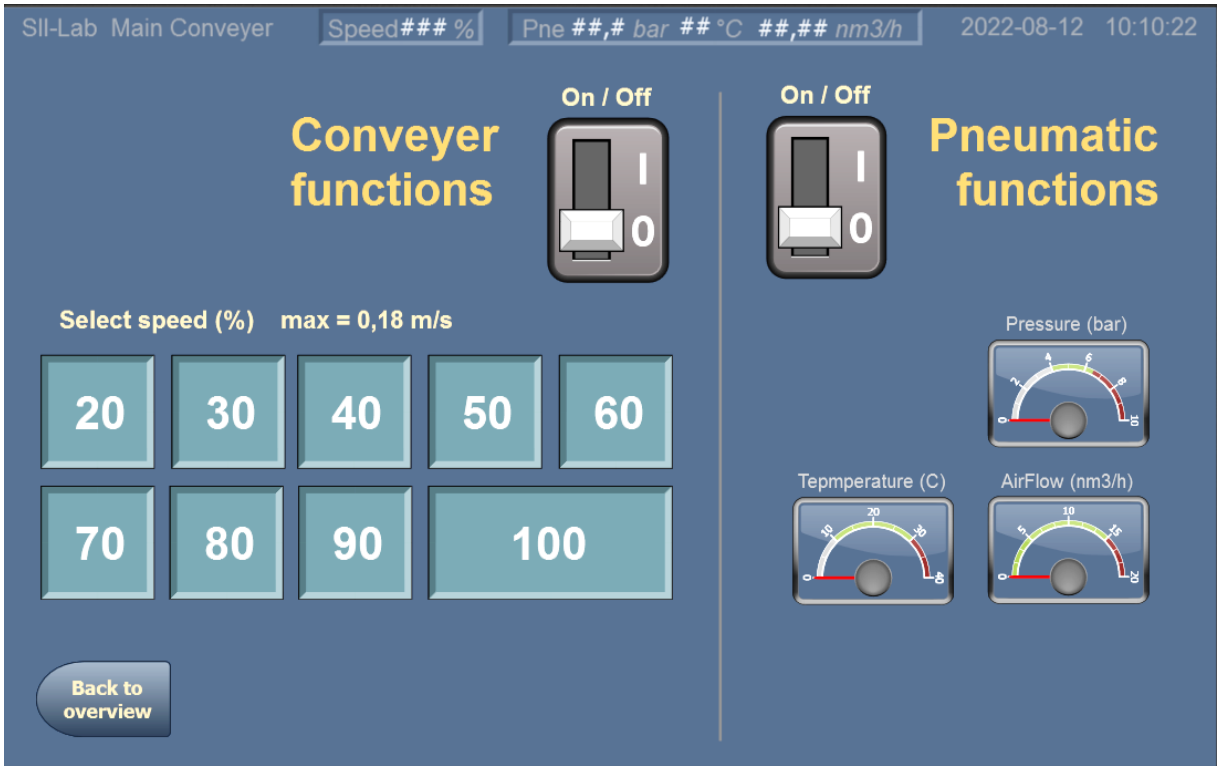
Sending the selected Mission to the controller

After selecting a Station and a Mission - press the button [*Send mission to controller*].
In the PopUp-dialog, as you can see below, press [Yes] for sending or press [No] for cancel



HMI screen

Conveyer and Pneumatic functions



Functions for conveyer

Start and stop conveyer

Push the icon [On/Off] to toggle TAG [*GVL_HMI.hmi_conveyer_on*]

Select speed for conveyer belt

Push on of the button [20]…[100] to select the % of maximum speed.
The maximum speed is around 0,18 m/s.
The selected speed is stored in TAG [*GVL_HMI.hmi_conveyer_speed*]

Functions for pneumatic

Open and close the valve for pneumatic airflow

Push the icon [On/Off] to toggle TAG [*GVL_HMI_hmi_pneumatic_pressure_on*]

Pressure (bar) meter:

Shows the air pressure in the system. A normal value shall be around **6 bar** and are stored in TAG [*GVL_Pneumatic.ifm_q3_pressure_sensor_value*]

Temperature (°C) meter:

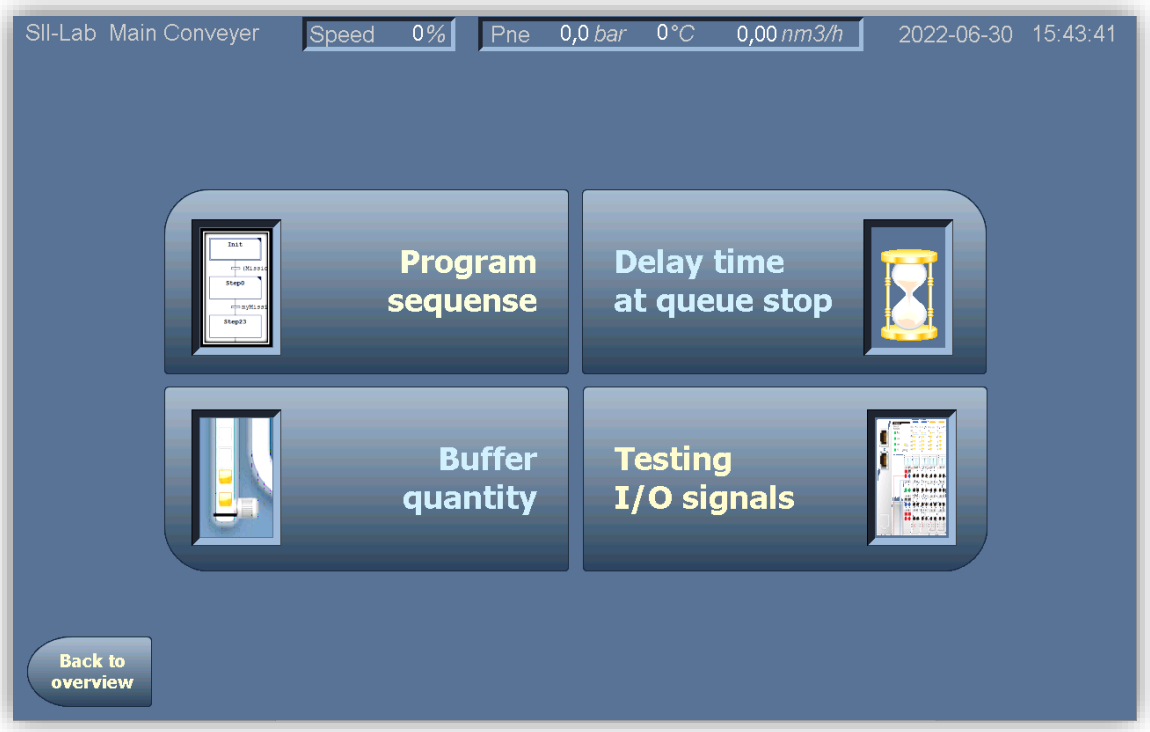
Shows the air temperature in the system. A normal value shall be around **20 °C** and are stored in TAG [*GVL_Pneumatic.ifm_sd5000_temperature_sensor_value*]

AirFlow (nm3/h) meter:

Shows the air flow in the system.
A normal maximum value shall not be higher than **15 nm3/h** and are stored in TAG [*GVL_Pneumatic.ifm_sd5000_flow_rate_sensor_value*]

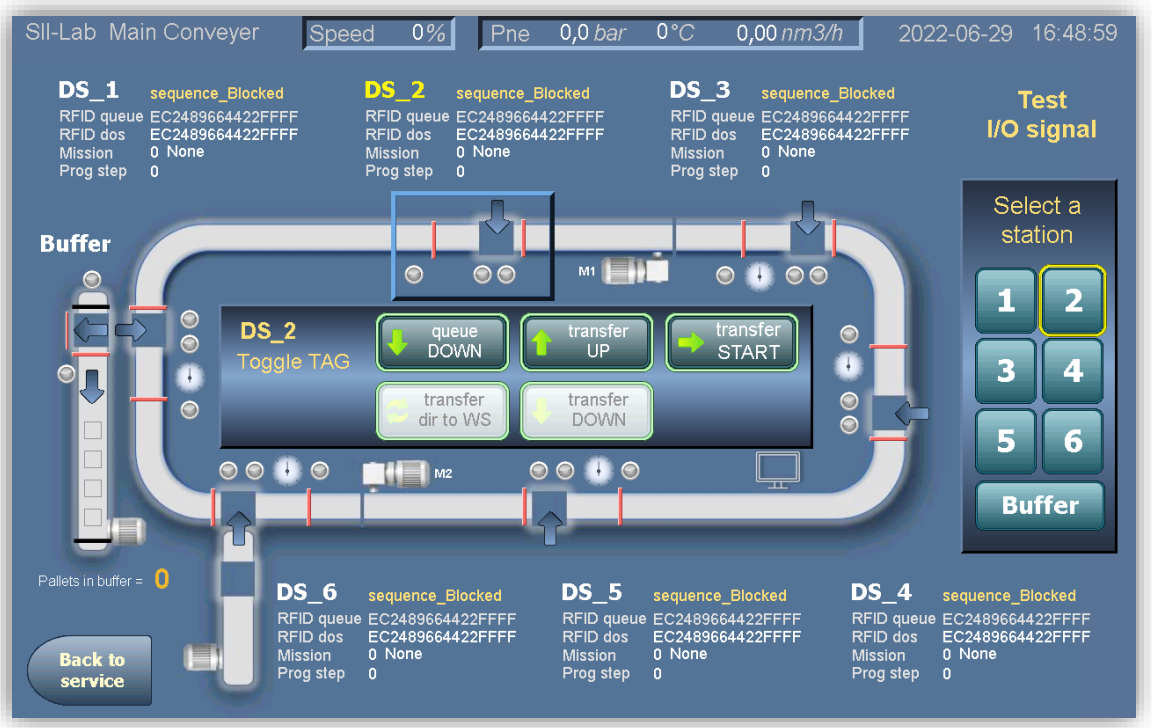
HMI screen

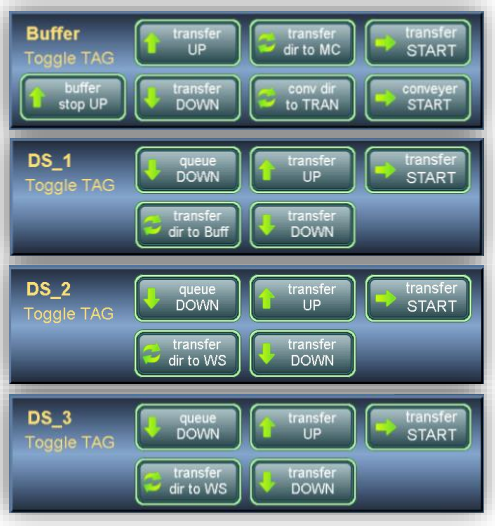
Menu



HMI screen

Test I/O signal





Available commands fore Buffer and DS1 – DS6

Explanation

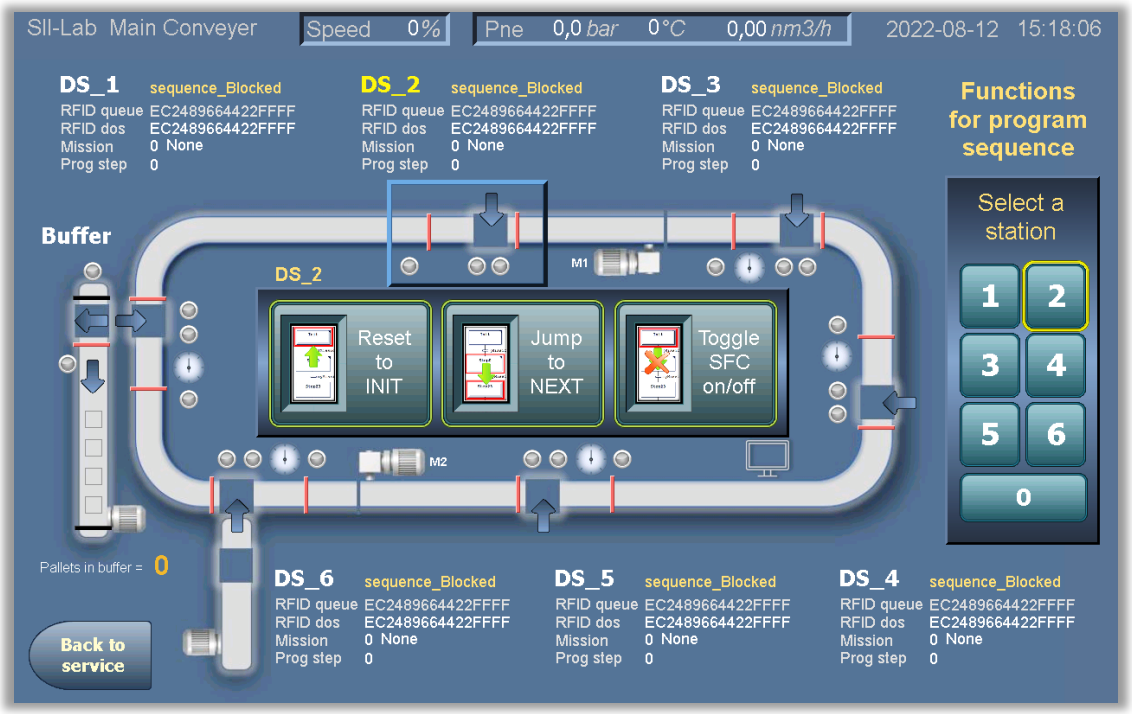
A green frame around the button means that the command is activated

You can't change the transfer direction when START is active

If you set the transfer position to Up or Down, you can't change to the opposite position.
To do that you need to Reset the activated command

HMI screen

Program sequence



Functions for program sequence (SFC)

Before start using the functions below – select a DS

Reset to INIT

Push the button to activate TAG [*GVL_DSx.MC_DSx_SFCInit*]
When the tag is high the program pointer jump to the Initial step.

Jump to NEXT:

Push the button to activate TAG [*GVL_DSx.MC_DSx_SFCTip*]
When the tag is high the program pointer jump directly to next step.

Toggle SFC on/off:

Push the button to toggle TAG [*GVL_DSx.MC_DSx_sequence_Blocked*]
When the tag is high the sequence is blocked in the transition condition direct after Init step.
This function are useful if you want to test I/O-signals.

HMI screen

Program sequence / Code 1(3)

```
public partial class Program_Sequence
{

    void Btn_ResToInit_MouseDown(System.Object sender, System.Windows.Forms.MouseEventArgs e)
    {
        switch (Globals.Tags.Selected_DoS.Value.Int)
        {
            case 0:
                break;

            case 1:
                Globals.Tags.Application_GVL_DS1_MC_DS1_SFCInit.SetTag();
                break;
            case 2:
                Globals.Tags.Application_GVL_DS2_MC_DS2_SFCInit.SetTag();
                break;
            case 3:
                Globals.Tags.Application_GVL_DS3_MC_DS3_SFCInit.SetTag();
                break;
            case 4:
                Globals.Tags.Application_GVL_DS4_MC_DS4_SFCInit.SetTag();
                break;
            case 5:
                Globals.Tags.Application_GVL_DS5_MC_DS5_SFCInit.SetTag();
                break;
            case 6:
                Globals.Tags.Application_GVL_DS6_MC_DS6_SFCInit.SetTag();
                break;
            default:
                break;
        }
    }


    void Btn_ResToInit_MouseUp(System.Object sender, System.Windows.Forms.MouseEventArgs e)
    {
        switch (Globals.Tags.Selected_DoS.Value.Int)
        {
            case 0:
                break;

            case 1:
                Globals.Tags.Application_GVL_DS1_MC_DS1_SFCInit.ResetTag();
                break;
            case 2:
                Globals.Tags.Application_GVL_DS2_MC_DS2_SFCInit.ResetTag();
                break;
            case 3:
                Globals.Tags.Application_GVL_DS3_MC_DS3_SFCInit.ResetTag();
                break;
            case 4:
                Globals.Tags.Application_GVL_DS4_MC_DS4_SFCInit.ResetTag();
                break;
            case 5:
                Globals.Tags.Application_GVL_DS5_MC_DS5_SFCInit.ResetTag();
                break;
            case 6:
                Globals.Tags.Application_GVL_DS6_MC_DS6_SFCInit.ResetTag();
                break;
            default:
                break;
        }
    }
```

HMI screen

Program sequence / Code 2(3)

```
void BtnJumpToNext_MouseDown(System.Object sender, System.Windows.Forms.MouseEventArgs e)
{
    switch (Globals.Tags.Selected_DoS.Value.Int)
    {
        case 0:
            break;

        case 1:
            Globals.Tags.Application_GVL_DS1_MC_DS1_SFCTip.SetTag();
            break;
        case 2:
            Globals.Tags.Application_GVL_DS2_MC_DS2_SFCTip.SetTag();
            break;
        case 3:
            Globals.Tags.Application_GVL_DS3_MC_DS3_SFCTip.SetTag();
            break;
        case 4:
            Globals.Tags.Application_GVL_DS4_MC_DS4_SFCTip.SetTag();
            break;
        case 5:
            Globals.Tags.Application_GVL_DS5_MC_DS5_SFCTip.SetTag();
            break;
        case 6:
            Globals.Tags.Application_GVL_DS6_MC_DS6_SFCTip.SetTag();
            break;
        default:
            break;
    }
}
```

```
void BtnJumpToNext_MouseUp(System.Object sender, System.Windows.Forms.MouseEventArgs e)
{
    switch (Globals.Tags.Selected_DoS.Value.Int)
    {
        case 0:
            break;

        case 1:
            Globals.Tags.Application_GVL_DS1_MC_DS1_SFCTip.ResetTag();
            break;
        case 2:
            Globals.Tags.Application_GVL_DS2_MC_DS2_SFCTip.ResetTag();
            break;
        case 3:
            Globals.Tags.Application_GVL_DS3_MC_DS3_SFCTip.ResetTag();
            break;
        case 4:
            Globals.Tags.Application_GVL_DS4_MC_DS4_SFCTip.ResetTag();
            break;
        case 5:
            Globals.Tags.Application_GVL_DS5_MC_DS5_SFCTip.ResetTag();
            break;
        case 6:
            Globals.Tags.Application_GVL_DS6_MC_DS6_SFCTip.ResetTag();
            break;
        default:
            break;
    }
}
```

HMI screen

Program sequence / Code 3(3)

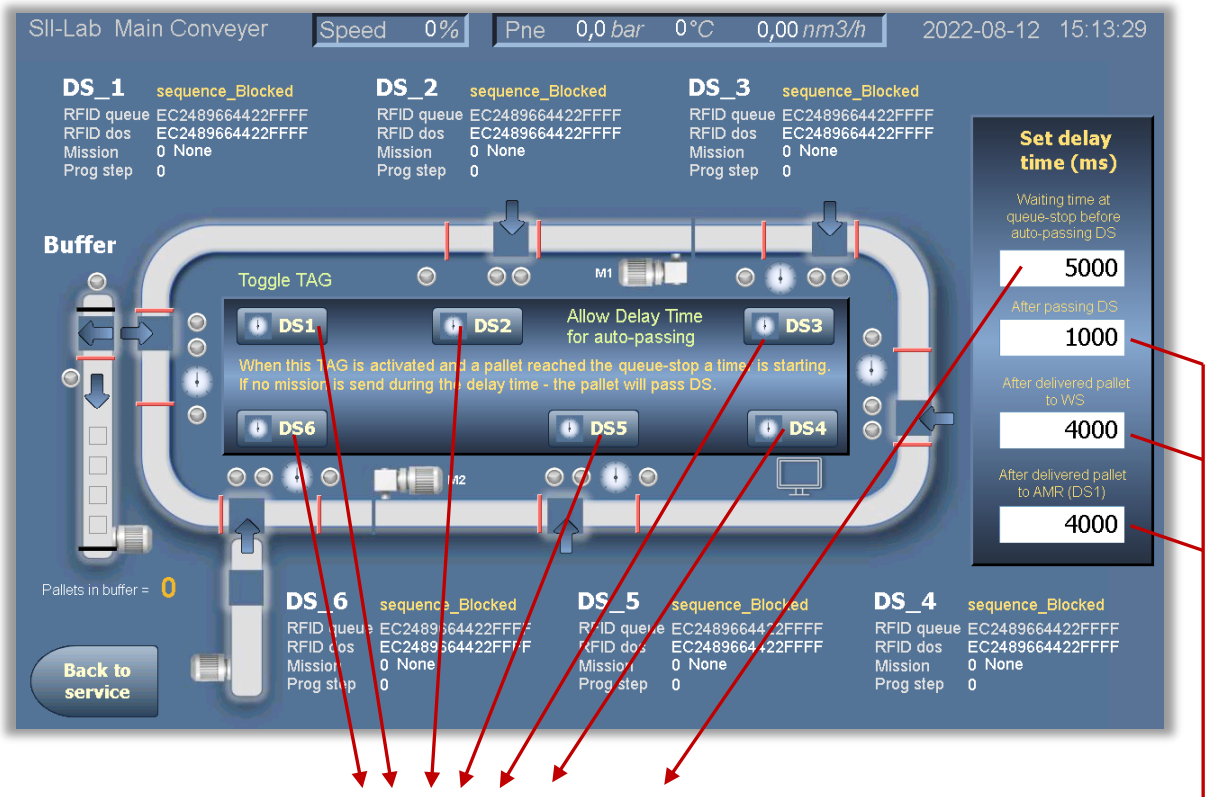
```
    void BtnToggleManuallyMode_Click(System.Object sender, System.EventArgs e)
    {
        switch (Globals.Tags.Selected_DoS.Value.Int)
        {
            case 0:
                break;

            case 1:
                Globals.Tags.Application_GVL_DS1_MC_DS1_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS1_MC_DS1_sequence_Blocked.Value;
                break;
            case 2:
                Globals.Tags.Application_GVL_DS2_MC_DS2_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS2_MC_DS2_sequence_Blocked.Value;
                break;
            case 3:
                Globals.Tags.Application_GVL_DS3_MC_DS3_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS3_MC_DS3_sequence_Blocked.Value;
                break;
            case 4:
                Globals.Tags.Application_GVL_DS4_MC_DS4_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS4_MC_DS4_sequence_Blocked.Value;
                break;
            case 5:
                Globals.Tags.Application_GVL_DS5_MC_DS5_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS5_MC_DS5_sequence_Blocked.Value;
                break;
            case 6:
                Globals.Tags.Application_GVL_DS6_MC_DS6_sequence_Blocked.Value = !Globals.Tags.Application_GVL_DS6_MC_DS6_sequence_Blocked.Value;
                break;
            default:
                break;
        }
    }

}
```

HMI screen

Delay functions

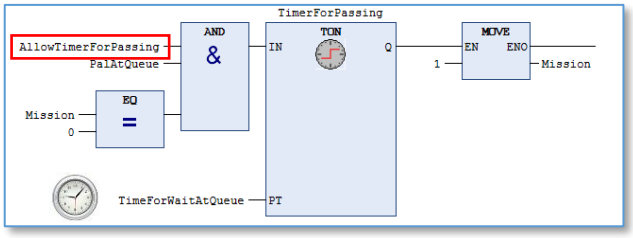


Allow delay functions for auto passing DS

To activate the auto function for passing a DS - set the following TAG to TRUE [*GVL_DSx_MC_DSx_Allow_Timer_For_Passing*].

If this TAG is TRUE and a pallet are at the queue position a TIMER start.

When the timer is >= [*GVL_HMI_hmi_TimeForWaitAtQueue*] the sequence for Passing DS starts. The picture below shows the program code for this function.



Other delay times that is possible to change:

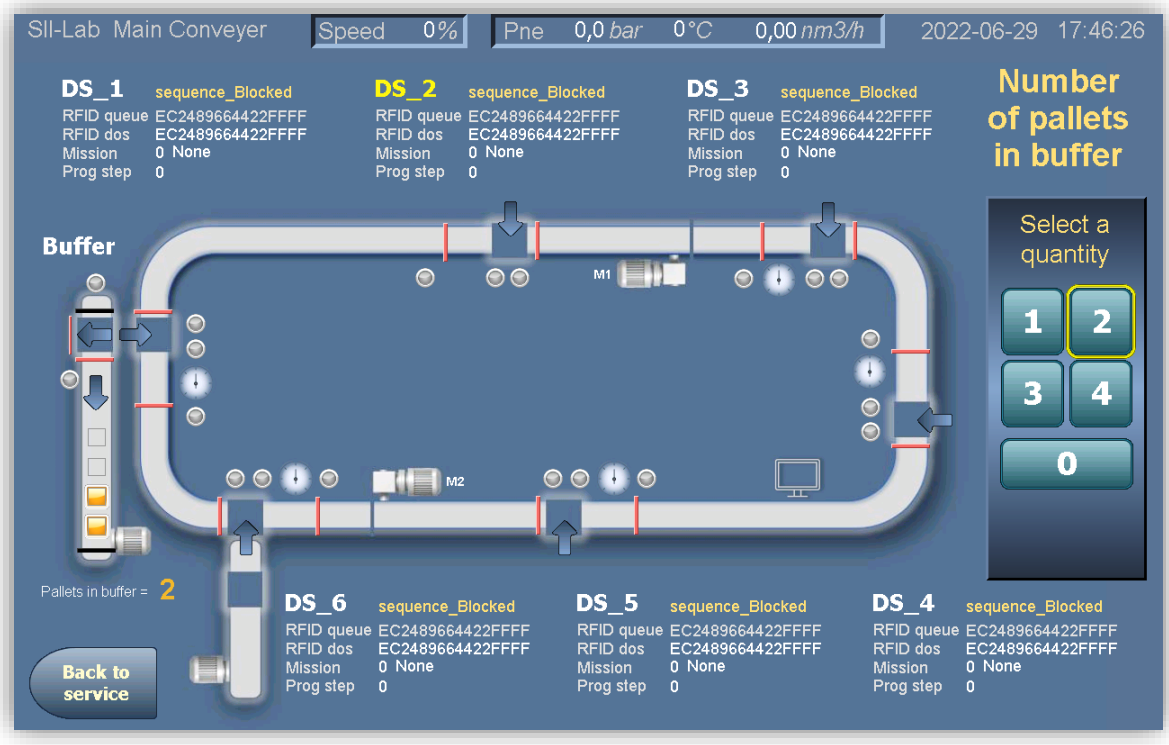
[*GVL_HMI_hmi_TimeForWaitAfterPassingDS*]

[*GVL_HMI_hmi_TimeForWaitAfterTransferdToWS*]

[*GVL_HMI_hmi_TimeForWaitAfterTransferdToAMR*]

HMI screen

Pallets in buffer



Set quantity of pallets in buffer

Push the button in the numeric pad to select the quantity of pallets in buffer DS1.
The value is stored in TAG [*GVL_DS1.MC_DS1_PalletsInBuffer*]

Appendix

GVL_HMI

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**hmi_conveyor_on**		BOOL	FALSE
VAR_GLOBAL	**hmi_conveyor_speed**		UINT	80
VAR_GLOBAL	**hmi_conveyor_ramp**		UINT	1000
VAR_GLOBAL	**hmi_reset_warning_malfunctioning**		BOOL	FALSE
VAR_GLOBAL	**hmi_pneumatic_pressure_on**		BOOL	
VAR_GLOBAL	**hmi_TimeForWaitAtQueue**		TIME	T#5000MS
VAR_GLOBAL	**hmi_TimeForWaitAfterPassingDS**		TIME	T#1000MS
VAR_GLOBAL	**hmi_TimeForWaitAfterTransferdToAMR**		TIME	T#4000MS
VAR_GLOBAL	**hmi_TimeForWaitAfterTransferdToWS**		TIME	T#4000MS

GVL_Pneumatic

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**pneumatic_pressure_on**		BOOL	TRUE
VAR_GLOBAL	**pneumatic_pressure_sensor_ok**		BOOL	
VAR_GLOBAL	**ifm_q3_pressure_sensor_value**		REAL	
VAR_GLOBAL	**ifm_q3_pressure_sensor_bytes**		ARRAY [0..1] OF BYTE	
VAR_GLOBAL	**ifm_sd5000_flow_rate_sensor_value**		REAL	
VAR_GLOBAL	**ifm_sd5000_flow_rate_sensor_bytes**		ARRAY [0..1] OF BYTE	
VAR_GLOBAL	**ifm_sd5000_temperature_sensor_value**		REAL	
VAR_GLOBAL	**ifm_sd5000_temperature_sensor_bytes**		ARRAY [0..1] OF BYTE	

GVL_Conveyer

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**conveyor_on**		BOOL	FALSE
VAR_GLOBAL	**conveyor_speed**		UINT	80
VAR_GLOBAL	**conveyor_ramp**		UINT	1000

Appendix

GVL_Conveyer_Motor

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**CM1_PO1_control_word**		WORD	
VAR_GLOBAL	**CM1_PO2_speed**		WORD	40960
VAR_GLOBAL	**CM1_PO3_ramp**		WORD	2000
VAR_GLOBAL	**CM1_PI1_controller_enabled**		BOOL	
VAR_GLOBAL	**CM1_PI1_inverter_ready**		BOOL	
VAR_GLOBAL	**CM1_PI1_PO_data_enabled**		BOOL	
VAR_GLOBAL	**CM1_PI1_malfunctioning_warning**		BOOL	
VAR_GLOBAL	**CM1_PI2_output_current**		WORD	
VAR_GLOBAL	**CM1_PI3_controller_enabled**		BOOL	
VAR_GLOBAL	**CM1_PI3_inverter_ready**		BOOL	
VAR_GLOBAL	**CM1_PI3_PO_data_enabled**		BOOL	
VAR_GLOBAL	**CM1_PI3_malfunctioning_warning**		BOOL	
VAR_GLOBAL	**CM1_PI3_brake**		BOOL	
VAR_GLOBAL	**CM1_PI3_O2_ready_for_operation**		BOOL	
VAR_GLOBAL	**CM1_PI3_CW**		BOOL	
VAR_GLOBAL	**CM1_PI3_CCW**		BOOL	
VAR_GLOBAL	**CM1_PI3_setpoint_f2**		BOOL	
VAR_GLOBAL	**CM2_PO1_control_word**		WORD	0
VAR_GLOBAL	**CM2_PO2_speed**		WORD	40960
VAR_GLOBAL	**CM2_PO3_ramp**		WORD	2000
VAR_GLOBAL	**CM2_PI1_controller_enabled**		BOOL	
VAR_GLOBAL	**CM2_PI1_inverter_ready**		BOOL	
VAR_GLOBAL	**CM2_PI1_PO_data_enabled**		BOOL	
VAR_GLOBAL	**CM2_PI1_malfunctioning_warning**		BOOL	
VAR_GLOBAL	**CM2_PI2_output_current**		WORD	
VAR_GLOBAL	**CM2_PI3_controller_enabled**		BOOL	
VAR_GLOBAL	**CM2_PI3_inverter_ready**		BOOL	
VAR_GLOBAL	**CM2_PI3_PO_data_enabled**		BOOL	
VAR_GLOBAL	**CM2_PI3_malfunctioning_warning**		BOOL	
VAR_GLOBAL	**CM2_PI3_brake**		BOOL	
VAR_GLOBAL	**CM2_PI3_O2_ready_for_operation**		BOOL	
VAR_GLOBAL	**CM2_PI3_CW**		BOOL	
VAR_GLOBAL	**CM2_PI3_CCW**		BOOL	
VAR_GLOBAL	**CM2_PI3_setpoint_f2**		BOOL	
VAR_GLOBAL	**conveyor_motors_warning_malfunctioning**		BOOL	FALSE
VAR_GLOBAL	**conveyor_motors_reset_warning_malfunctioning**		BOOL	FALSE

Appendix

GVL_RFID

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**MC_DOS1_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS1_RFID_String**		STRING	''
VAR_GLOBAL	**MC_DOS2_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS2_RFID_String**		STRING	''
VAR_GLOBAL	**MC_DOS3_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS3_RFID_String**		STRING	''
VAR_GLOBAL	**MC_DOS4_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS4_RFID_String**		STRING	''
VAR_GLOBAL	**MC_DOS5_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS5_RFID_String**		STRING	''
VAR_GLOBAL	**MC_DOS6_RFID_ByteArray**		ARRAY[0..7] OF BYTE	
VAR_GLOBAL	**MC_DOS6_RFID_String**		STRING	''

Appendix

GVL_DS1

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	MC_DS1_pallet_sensor_1		BOOL	
VAR_GLOBAL	MC_DS1_pallet_sensor_2		BOOL	
VAR_GLOBAL	MC_DS1_transfer_elevator_sensor_up		BOOL	
VAR_GLOBAL	MC_DS1_transfer_elevator_sensor_down		BOOL	
VAR_GLOBAL	MC_DS1_pneumatic_pressure_sensor_ok		BOOL	
VAR_GLOBAL	MC_DS1_pallet_queue_stop_sensor		BOOL	
VAR_GLOBAL	MC_DS1_queue_stop_down		BOOL	FALSE
VAR_GLOBAL	MC_DS1_transfer_elevator_down		BOOL	FALSE
VAR_GLOBAL	MC_DS1_transfer_elevator_up		BOOL	FALSE
VAR_GLOBAL	MC_DS1_transfer_on		BOOL	FALSE
VAR_GLOBAL	MC_DS1_transfer_direction_from_mc		BOOL	FALSE
VAR_GLOBAL	MC_DS1_pneumatic_pressure_on		BOOL	FALSE
VAR_GLOBAL	MC_DS1_sequence_Blocked		BOOL	FALSE
VAR_GLOBAL	MC_DS1_Allow_Timer_For_Passing		BOOL	TRUE
VAR_GLOBAL	MC_DS1_SFCInit		BOOL	
VAR_GLOBAL	MC_DS1_SFCTip		BOOL	
VAR_GLOBAL	MC_DS1_SFCCurrentStep		STRING	
VAR_GLOBAL	MC_DS1_Mission		USINT	
VAR_GLOBAL	MC_DS1_myMission		USINT	
VAR_GLOBAL	MC_DS1_myRFID_OTW		STRING	

GVL_DS1_Buffer

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	MC_DS1_buffer_stop_sensor		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_elevator_sensor_up		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_elevator_sensor_down		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_stop_up		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_elevator_up		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_elevator_down		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_direction_to_mc		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_on		BOOL	
VAR_GLOBAL	MC_DS1_buffer_conveyor_direction_to_transfer		BOOL	
VAR_GLOBAL	MC_DS1_buffer_conveyor_on		BOOL	
VAR_GLOBAL	MC_DS1_buffer_transfer_stop_sensor		BOOL	
VAR_GLOBAL	MC_DS1_PalletsInBuffer		USINT	

Appendix

GVL_DS2

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	MC_DS2_pallet_sensor_1		BOOL	
VAR_GLOBAL	MC_DS2_pallet_sensor_2		BOOL	
VAR_GLOBAL	MC_DS2_transfer_elevator_sensor_up		BOOL	
VAR_GLOBAL	MC_DS2_transfer_elevator_sensor_down		BOOL	
VAR_GLOBAL	MC_DS2_pallet_queue_stop_sensor		BOOL	
VAR_GLOBAL	MC_DS2_queue_stop_down		BOOL	
VAR_GLOBAL	MC_DS2_transfer_elevator_down		BOOL	
VAR_GLOBAL	MC_DS2_transfer_elevator_up		BOOL	
VAR_GLOBAL	MC_DS2_transfer_on		BOOL	
VAR_GLOBAL	MC_DS2_transfer_direction_from_mc		BOOL	
VAR_GLOBAL	MC_DS2_mission5_completed		BOOL	
VAR_GLOBAL	MC_DS2_sequence_Blocked		BOOL	FALSE
VAR_GLOBAL	MC_DS2_Allow_Timer_For_Passing		BOOL	TRUE
VAR_GLOBAL	MC_DS2_SFCInit		BOOL	
VAR_GLOBAL	MC_DS2_SFCTip		BOOL	
VAR_GLOBAL	MC_DS2_SFCCurrentStep		STRING	
VAR_GLOBAL	MC_DS2_Mission		USINT	
VAR_GLOBAL	MC_DS2_myMission		USINT	
VAR_GLOBAL	MC_DS2_myRFID_OTW		STRING	

GVL_DS3

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	MC_DS3_pallet_sensor_1		BOOL	
VAR_GLOBAL	MC_DS3_pallet_sensor_2		BOOL	
VAR_GLOBAL	MC_DS3_transfer_elevator_sensor_up		BOOL	
VAR_GLOBAL	MC_DS3_transfer_elevator_sensor_down		BOOL	
VAR_GLOBAL	MC_DS3_pallet_queue_stop_sensor		BOOL	
VAR_GLOBAL	MC_DS3_queue_stop_down		BOOL	
VAR_GLOBAL	MC_DS3_transfer_elevator_down		BOOL	
VAR_GLOBAL	MC_DS3_transfer_elevator_up		BOOL	
VAR_GLOBAL	MC_DS3_transfer_on		BOOL	
VAR_GLOBAL	MC_DS3_transfer_direction_from_mc		BOOL	
VAR_GLOBAL	MC_DS3_mission5_completed		BOOL	
VAR_GLOBAL	MC_DS3_sequence_Blocked		BOOL	FALSE
VAR_GLOBAL	MC_DS3_Allow_Timer_For_Passing		BOOL	TRUE
VAR_GLOBAL	MC_DS3_SFCInit		BOOL	
VAR_GLOBAL	MC_DS3_SFCTip		BOOL	
VAR_GLOBAL	MC_DS3_SFCCurrentStep		STRING	
VAR_GLOBAL	MC_DS3_Mission		USINT	
VAR_GLOBAL	MC_DS3_myMission		USINT	
VAR_GLOBAL	MC_DS3_myRFID_OTW		STRING	

Appendix

GVL_DS4

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**MC_DS4_pallet_sensor_1**		BOOL	
VAR_GLOBAL	**MC_DS4_pallet_sensor_2**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_elevator_sensor_up**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_elevator_sensor_down**		BOOL	
VAR_GLOBAL	**MC_DS4_pallet_queue_stop_sensor**		BOOL	
VAR_GLOBAL	**MC_DS4_queue_stop_down**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_elevator_down**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_elevator_up**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_on**		BOOL	
VAR_GLOBAL	**MC_DS4_transfer_direction_from_mc**		BOOL	
VAR_GLOBAL	**MC_DS4_mission5_completed**		BOOL	
VAR_GLOBAL	**MC_DS4_sequence_Blocked**		BOOL	FALSE
VAR_GLOBAL	**MC_DS4_Allow_Timer_For_Passing**		BOOL	TRUE
VAR_GLOBAL	**MC_DS4_SFCInit**		BOOL	
VAR_GLOBAL	**MC_DS4_SFCTip**		BOOL	
VAR_GLOBAL	**MC_DS4_SFCCurrentStep**		STRING	
VAR_GLOBAL	**MC_DS4_Mission**		USINT	
VAR_GLOBAL	**MC_DS4_myMission**		USINT	
VAR_GLOBAL	**MC_DS4_myRFID_OTW**		STRING	

GVL_DS5

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**MC_DS5_pallet_sensor_1**		BOOL	
VAR_GLOBAL	**MC_DS5_pallet_sensor_2**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_elevator_sensor_up**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_elevator_sensor_down**		BOOL	
VAR_GLOBAL	**MC_DS5_pallet_queue_stop_sensor**		BOOL	
VAR_GLOBAL	**MC_DS5_queue_stop_down**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_elevator_down**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_elevator_up**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_on**		BOOL	
VAR_GLOBAL	**MC_DS5_transfer_direction_from_mc**		BOOL	
VAR_GLOBAL	**MC_DS5_mission5_completed**		BOOL	
VAR_GLOBAL	**MC_DS5_sequence_Blocked**		BOOL	FALSE
VAR_GLOBAL	**MC_DS5_Allow_Timer_For_Passing**		BOOL	TRUE
VAR_GLOBAL	**MC_DS5_SFCInit**		BOOL	
VAR_GLOBAL	**MC_DS5_SFCTip**		BOOL	
VAR_GLOBAL	**MC_DS5_SFCCurrentStep**		STRING	
VAR_GLOBAL	**MC_DS5_Mission**		USINT	
VAR_GLOBAL	**MC_DS5_myMission**		USINT	
VAR_GLOBAL	**MC_DS5_myRFID_OTW**		STRING	

Appendix

GVL_DS6

Scope	Name	Address	Data type	Initialization
VAR_GLOBAL	**MC_DS6_pallet_sensor_1**		BOOL	
VAR_GLOBAL	**MC_DS6_pallet_sensor_2**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_elevator_sensor_up**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_elevator_sensor_down**		BOOL	
VAR_GLOBAL	**MC_DS6_pallet_queue_stop_sensor**		BOOL	
VAR_GLOBAL	**MC_DS6_queue_stop_down**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_elevator_down**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_elevator_up**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_on**		BOOL	
VAR_GLOBAL	**MC_DS6_transfer_direction_from_mc**		BOOL	
VAR_GLOBAL	**MC_DS6_sequence_Blocked**		BOOL	FALSE
VAR_GLOBAL	**MC_DS6_Allow_Timer_For_Passing**		BOOL	TRUE
VAR_GLOBAL	**MC_DS6_SFCInit**		BOOL	
VAR_GLOBAL	**MC_DS6_SFCTip**		BOOL	
VAR_GLOBAL	**MC_DS6_SFCCurrentStep**		STRING	
VAR_GLOBAL	**MC_DS6_Mission**		USINT	
VAR_GLOBAL	**MC_DS6_myMission**		USINT	
VAR_GLOBAL	**MC_DS6_myRFID_OTW**		STRING	